Crack Detection

EE397 Technical Report

Group 5:

Patricia Ehigiamusoe

Shakirat Olayode

Md Rayhan Mia

Facilitator: Vahid Fakhari

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Department of Electronic Engineering

Maynooth University

ABSTRACT

The aim of this report is to present a possible solution to a societal issue, aligned with the Sustainable Development Goals (SDGs). The solution must be well researched and extensively tested to determine how well is addresses a certain issue or part of an issue.

Sustainable infrastructure is a key part of any type of construction. From physical buildings to web applications, every developer works to have infrastructure that is easily maintainable and will not fail unexpectedly. One aspect of maintenance is testing for irregularities. This should be done during development as well as on a regular basis after development is complete.

Looking at the area of infrastructure sustainability, we decided on a crack detection device. Such a device detects surface level cracks within a degree of accuracy. This report details our findings in this area as well as our efforts to create a crack detecting device.

Throughout this report, the research behind a crack detection device and the construction of our device will be demonstrated.

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1. INTRODUCTION

Crack, break, fracture, split. These are not words we want to hear in any building or manufacturing process. Nowadays, we are so used to seeing cracks all around us. For example, in our homes, in buildings, and on bridges. Some are big, but many are so small we don't think twice about them. But these miniature fractures can lead to major issues if left unchecked and the first step in addressing any problem is being aware of the problem. This is our motivation for developing a crack detection device. We are aware that many versions of similar devices exist. However, they are usually expensive and therefore not accessible for projects with a smaller budget. By developing a device at a lower cost and that is simple to use, we hope to make crack detection more accessible.

This aims of this project align with the 9th Sustainable Development Goal (SDG), Industry, Innovation, and Infrastructure. We will reflect on the social and environmental impact cracks and fractures have on infrastructure.

In this report, we will discuss the theoretical aspect of crack detection as well as the technical work carried out to create a prototype device.

2. REPORT LAYOUT

The rest of this report is structured as follows:

- In section 3, a detailed review of the crack detection field is investigated. Examples of existing devices are looked at. This provides a strong basis for the project to be built on.
- Section 4 covers the ethical considerations surrounding this project, including a related Sustainable Development Goal (SDG), Industry, Innovation and Infrastructure. This section aims to examine the role of maintenance and sustainability in society.
- In section 5, customer input is gathered via survey. An analysis of public awareness and opinion of infrastructure issues and their solutions provide valuable insights into the necessity of this project.
- In section 6, the development of our first prototype is discussed. This includes sensor testing and the design of an external cover.
- Section 7 presents customer feedback of the prototype. A new post-development survey is compared to the initial survey carried out at the beginning of the project.
- Finally, section 8 highlights the main findings of this project in a conclusion. Possible further works are also explored in this section.

3. LITERATURE REVIEW

Looking at available crack detection devices will further broaden our understanding of crack detection technology and how such a device would need to be developed. This review will explore different types of crack detection which will help us decide the best option to use for our project.

3.1. Concrete Ultrasonic Pulse Flaw Detector

The Concrete Ultrasonic Pulse Flaw Detector for Crack Depth and Width is a device sold by GAOTek [1]. It is advertised as an “advanced ultrasonic flaw detector with 1-12,000mm range”. Some of the parts it consists of are an LCD, a crack width probe, a crack depth gauge scale and couplant. The device works by reading the data with the transducer (probe) and displaying it on screen. A couplant (gel like substance) is used to improve the transmission of ultrasonic waves between the probe and material being tested [2]. There is no price listed for this product, there is only an option on their website to request a quote. However, we can imagine the price point of a device like this would be high.

Figure 1: Concrete Ultrasonic Pulse Flaw Detector for Crack Depth and Width

Figure 2: Parts included in Concrete Ultrasonic Pulse Flaw Detector

3.2.Eddy Current Testing (ECT)

Crack detecting products developed by Eddyfi Technologies use a technique called eddy current testing (ECT) [3]. ECT involves a passing alternating current through a copper wire which creates a magnetic field around the wire [4]. The current in the wire and the magnetic field oscillate at the same frequency. If the coil is brought near other conductive materials, the opposing currents create eddy currents. Defects are detected by analysing the change in the magnetic fields. They state that is type of testing offers “fast inspection of surface-breaking cracks”.

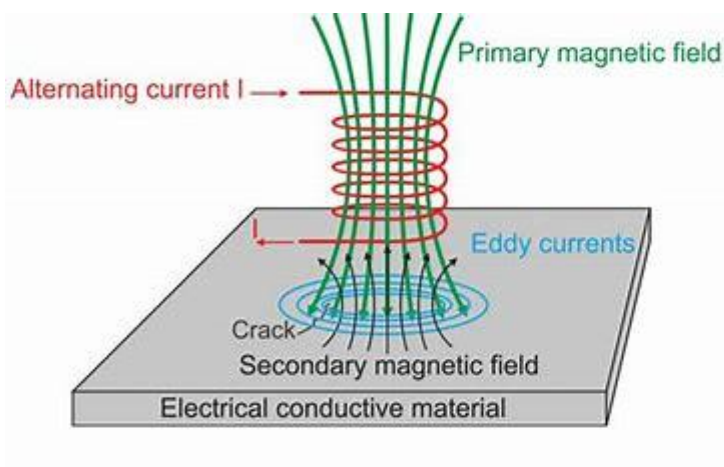

Figure 3: Eddy Current Testing (ECT)

By researching some available methods of crack detection, we have settled on using an ultrasonic sensor for our project. The straight-forward approach of the sensor is suitable for this project. The Arduino ultrasonic sensor can be easily coded and tested with a compatible microcontroller.

Ultrasonic crack detection which is a method of non-destructive testing (NDT) [5]. NDT is a form of testing that does not impact the test material, unlike destructive testing. Destructive testing evaluates the test material properties by forcing the properties (e.g. tensile strength) to reach their limit.

4. ETHICAL CONSIDERATIONS

4.1.Sustainable Development Goals (SDGs)

Part of the given criterion for this project was to provide a potential solution to a community-based problem. The Sustainable Development Goals (SDGs) are a set of seventeen global goals arranged by the United Nations [6]. These goals seek to tackle various global challenges, like No Poverty, Zero Hunger, Quality Education, and many more. The SDGs act as a “universal call to action to end poverty, protect the planet, and ensure prosperity for all” [6].

For this assignment, we choose to consider the 9th Sustainable Development Goal, ’Industry, Innovation, and Infrastructure’. The aim of this goal is to “build resilient infrastructure, promote inclusive and sustainable industrialisation and foster innovation” [7]. A crack detector device can be used to test the integrity of materials before, during and after construction.

4.2. Societal Problems

A societal problem is defined as “a condition or situation that negatively affects a significant number of people in society” [8]. Poor infrastructure conditions is an issue many people face. Many of the buildings we live and work in are left in poor conditions which can be a danger to people in them. In extreme cases the buildings can become unusable and left abandoned. This is a problem that can stem from different issues including the presence of small cracks or faults which worsen over time. One correct procedure to deal with such an issue is to regularly check the state of a building with a device like a crack detector to identify areas before they intensify.

With our device a user will be able to detect the issue early, limiting the potential permanent damages. The simple design and functionality make it easy to use so it's use is not restricted to just professionals.

4.3. Environment

Cracks and other structural defects play a major role in our environment today, more than we realise. The structural defects in buildings and other establishments have far-reaching impacts on the societal and economic infrastructure of the world. It affects public safety, service continuity and long-term development. The lack of considerations regarding the severity of "minor cracks" can lead to major safety issues and injuries or in some extreme cases death.

Examples of such cases include the collapse of the eight-storey Rana Plaza Factory in Bangladesh on 23rd April 2013. The building was evacuated due to the appearing of cracks in the walls the day before this mishap. However, as a result of failure to recognise the severity of the situation, workers were permitted back inside the following day. Disastrously, the building collapsed and resulted in significant loss of life [9]. The presence of these external cracks indicate that the structural integrity of the building had been compromised prior to this incident. If the cracks were detected earlier, the situation could have potentially been avoided.

There are multiple others severe cases similar to the Rana Plaza Factory case. For example, a sinkhole in Thailand in September 2025 [10] and the collapse of a bridge in China in July 2024 [11]. All these fatalities developed from construction oversights, including inadequate planning and insufficient long-term structural considerations. Over time, the emergence of cracks and structural decay led to fatal outcomes.

A possible solution to problems like this is a crack detector device. With this device, it is simple to check for cracks in most structures. This device is one step closer to a safer living environment.

5. CUSTOMER INPUT

When developing a product for public use, it is vital to consider who the end user will be. The target audience we looked at included professionals working in fields like construction, manufacturing and quality control, as well as people who engage in advanced home projects. We wanted a device that is suitable for both these groups. We first organised a collection of questions to ask people who might use a device like this. When choosing the questions, we wanted to assess the participants knowledge of crack detection and explore what features they would like in the device. We conducted a short survey to inform decisions later on in the project. The results of this survey are detailed below.

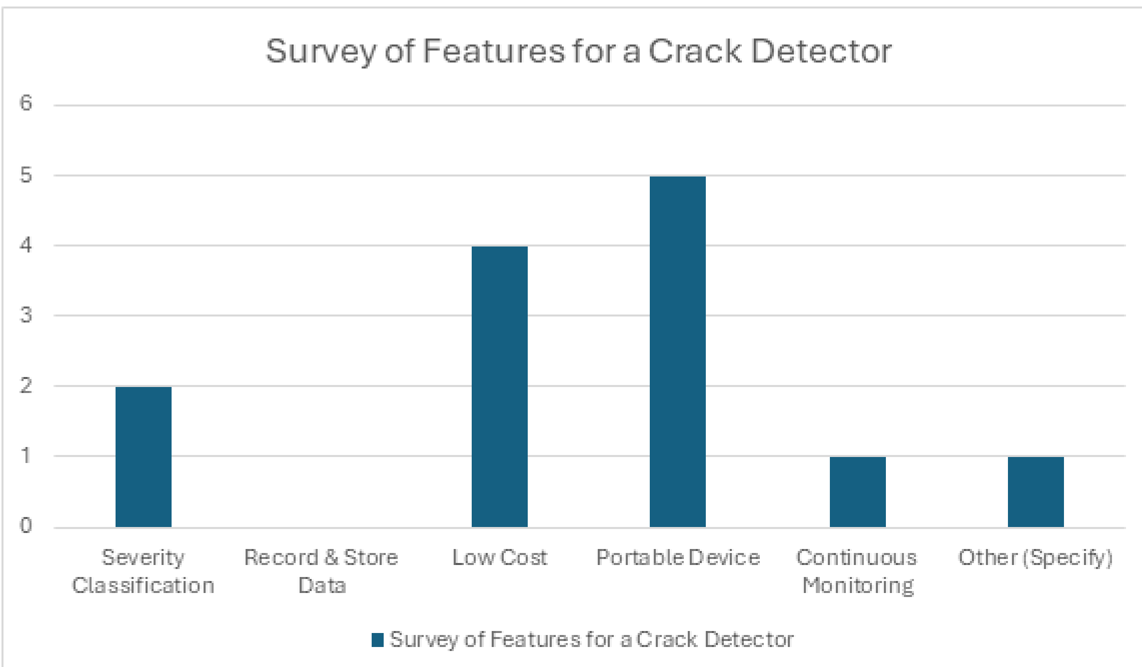

Figure 1: Desired Features in a Crack Detector from Survey

The graph above (Figure 1) shows the desired features chosen by the participants. They were given multiple options and could pick as many as they like. The most popular choice was a portable device. Another common choice was low cost. The customer input gathered from this survey defined the design of our prototype. We aimed to make the device small, easy to handle, portable and at a low price since those were the features chosen by most of the participants.

Outside of device features, 80% of respondents said they had heard of crack detecting tools. All participants said they would trust a device to detect cracks. This represents a level of trust and expectation in modern technology. People expect devices to be accurate and function well. This is something we considered when creating a prototype.

6. PROTOTYPE DEVELOPMENT

6.1. Sensors

Sensors are input devices that can monitor physical signals and detect different types of changes in the environment. They are crucial inspecting the behaviour of a system and detecting unanticipated actions or failures in a system [12]. There are multiple different types of sensors but for this project we settled on using an ultrasonic sensor.

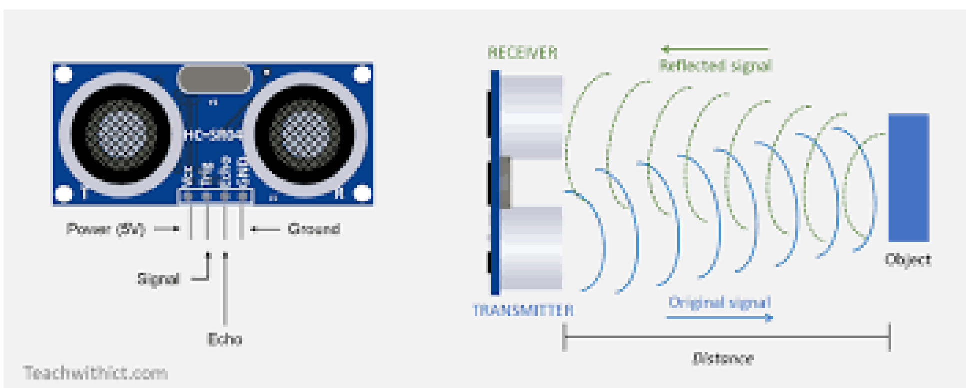

Figure 2: Ultrasonic Sensor

For the circuit, the HC-SR04 Ultrasonic Sensor was used. This sensor measures the distance by emitting high frequency sound waves and detecting their echo's. The datasheet of the sensor claims the sensor can detect objects in the range of 2cm – 400cm [13]. By placing this sensor opposite a surface within the range of detection, it can detect cracks using the sound waves.

6.2. Circuit Design

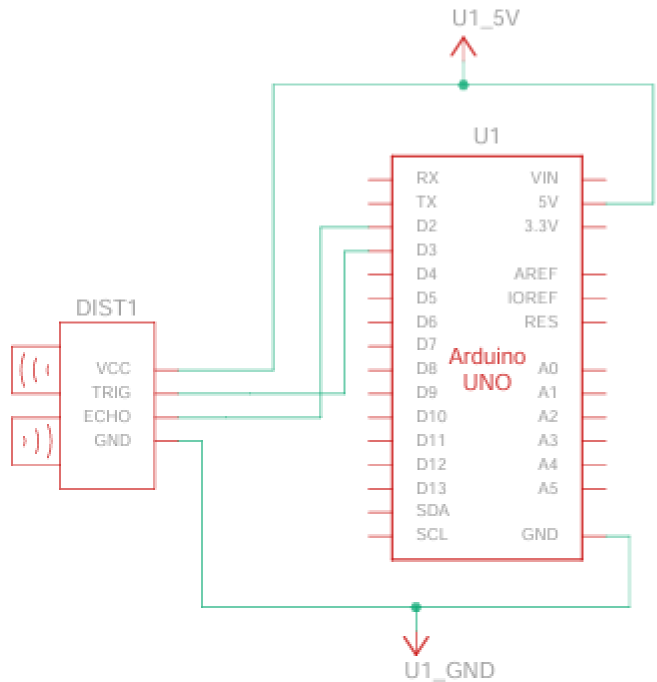

Figure 3: Circuit Wiring of Ultrasonic Sensor with Arduino UNO

The circuit components were composed of an ultrasonic sensor and an Arduino UNO microcontroller board. The code was written using the Arduino IDE [14]. The IDE provides a serial monitor and serial plotter. The monitor was used to check the printed distance values. The serial plotter was used to make a live graph of the data read from the ultrasonic sensor.

6.3. Testing Ultrasonic Sensor Accuracy

Since choosing to use an ultrasonic sensor, we needed to confirm the sensor we got operates as expected. The datasheet for the HC-SR04 Ultrasonic Sensor claims the operating range of the sensor is approximately 2cm to 400cm with ±3mm accuracy [15]. This test focuses on the lower side of the range. As the device is small, it will be limited to detecting smaller cracks. Therefore, the full range of the sensor will not be used. The details of the test to inspect the datasheet values are explained below.

Equipment Needed:

- Ultrasonic sensor
- Tape measure
- Arduino Uno
- Computer with Arduino IDE installed
- Object for sensor to detect

Method

- Choose a distance, measure the distance with a tape measure and record the value.
- Place an object at one end of the measured distance and the ultrasonic sensor at the other end.
- Run the code in Appendix C for the sensor to print values to the serial monitor and record the value.
- Comparing the actual distance to the value from the ultrasonic sensor distance to get the difference [Equation 1].

$$Difference = Actual\ distance - Ultrasonic\ sensor\ distance \qquad \text{[Equation 1]}$$

Results

Actual Distance (cm)	Ultrasonic Sensor Distance (cm)	Difference (cm)
2	1.99	0.01
8	7.79	0.21
0.5	0.49	0.01
1	1.04	-0.04
2.5	2.35	0.15
3	3.01	0.01
3.6	3.59	0.01

Table 1: Ultrasonic Sensor Accuracy Test Results

The result of the test shown in the table above (table 1), show the sensor is quite accurate. The biggest difference was 0.21cm, which converts to 2.1mm. This value is within the $\pm$3mm error from the datasheet. It can be concluded that the ultrasonic sensor operates as expected.

6.4. Simulations

The below figures (figure 4 and figure 5) are graphs of the distance measured when a crack is not present and when it is present. It is displayed using the serial plotter tool in the Arduino IDE. The x-axis is the time in microseconds. The y-axis is the measured distance in mm.

Figure 4: Graph showing no crack detected

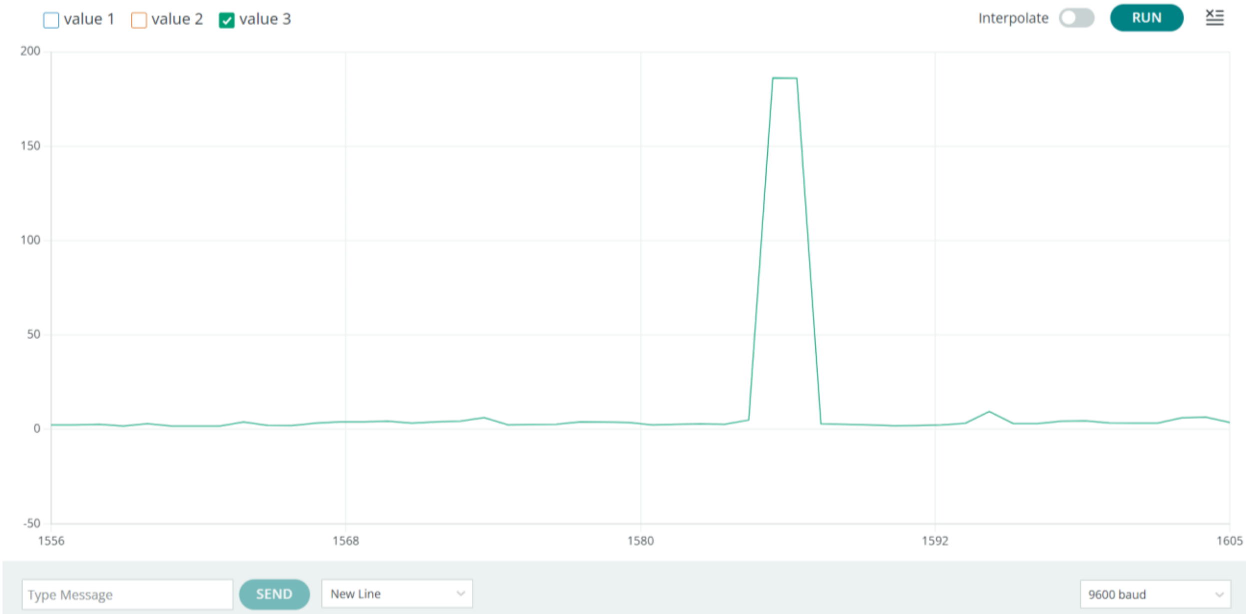

Figure 5: Graph showing crack detected

6.5.Enclosure Design

The enclosure for the Arduino Uno and HC-SR04 ultrasonic sensor used in the crack detection prototype was designed using Fusion 360. Factors influencing the enclosure design included the mounting geometries of the Arduino Uno, the automatic manufacturability of the enclosure, and the results of the crack detection system pre-development survey. Survey respondents stressed the need for a portable, low-cost, and user-friendly system. The housing design of the electronics began with a Google Sketchup workflow of an Arduino Uno dimension drawing (shown below) with modifications to accommodate the ultrasonic sensor.

Design Methodology

Using the two-point rectangle and dimensioned mounting holes of the external footprint of the Arduino Uno, the enclosure geometry was created. Prior to shelling, internal clearance (wiggle room) must be included. This eases assembly and prevents over-constraining or excessive friction as the Arduino board is slid into place. The enclosure body was then extruded at a fixed height and automatically shelled, resulting in a uniform component thickness. The body was then split in half at the mid-plane automatically and snap-fitted together with printed mounting pins and mounting holes.

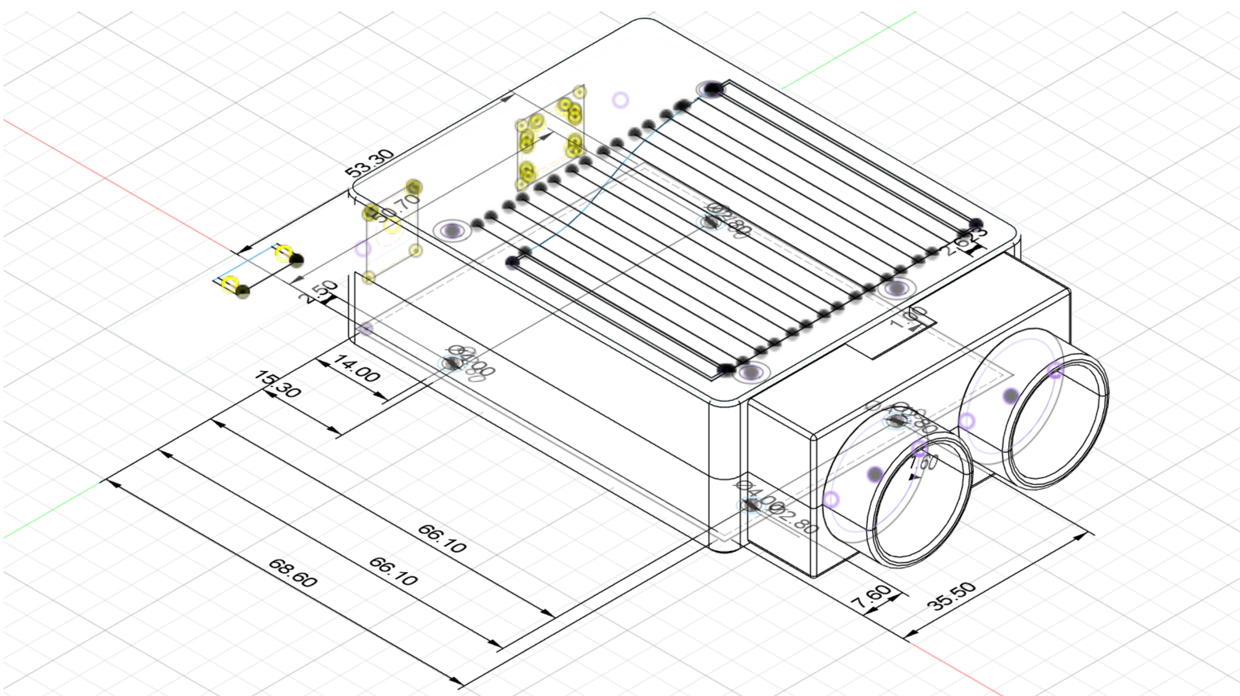

Figure 4: Dimensioned CAD model of the whole enclosure

Main Enclosure Body Dimensions

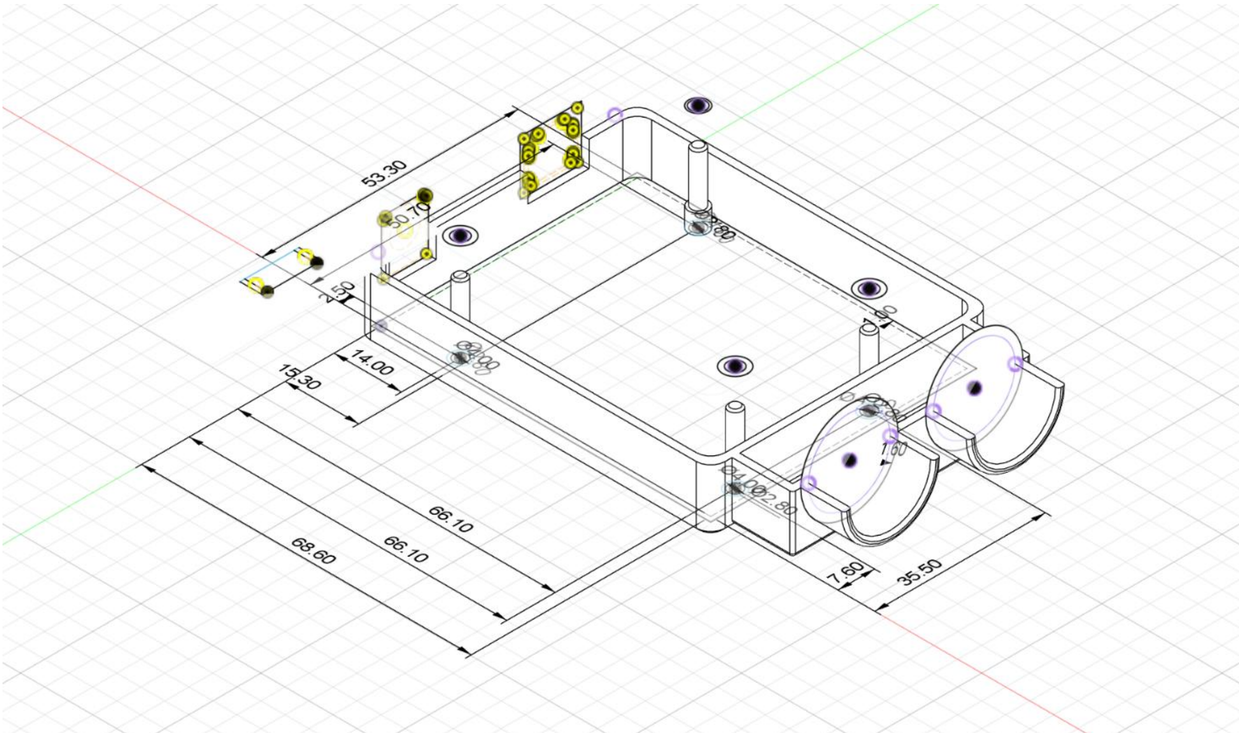

Figure 5: Dimensioned CAD model of the bottom part of the enclosure

External Dimensions

- Overall enclosure length: 68.60mm
- Overall enclosure width/height: 53.30mm
- Enclosure height (extrusion): 22.00mm

These dimensions are for the outer enclosure body and exclude the ultrasonic sensor housing.

Wall Thickness and Clearance

- Wall thickness (shell): 1.6mm
- Internal clearance (wiggle room): 1.0mm

The 1.0 mm clearance was accounted for by offsetting the Arduino reference geometry prior to the shelling operation. This also accounts for manufacturing tolerances, 3D printing.

shrinkage and ease of insertion and removal of the board. The wall thickness of 1.6 mm is equivalent to two shells around the perimeter in a 0.4 mm nozzle, making the wall strong while minimising material.

Mounting Pillars and Arduino Interface

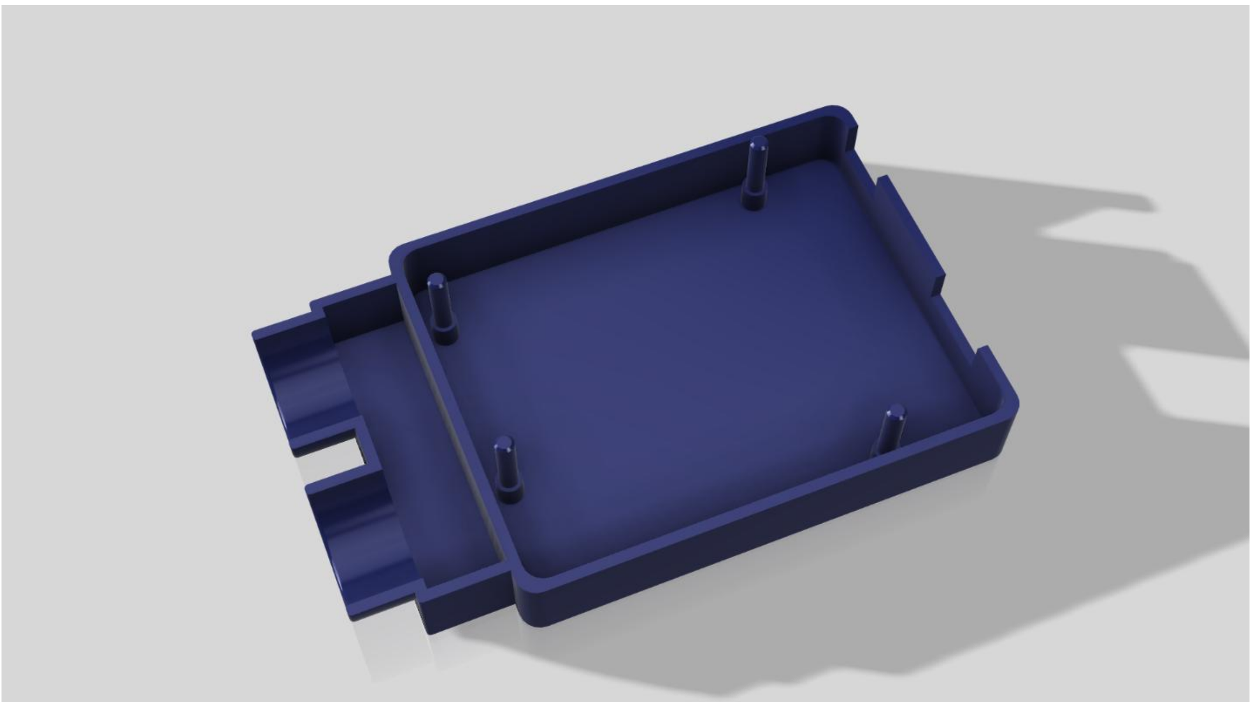

Figure 6: 3D model of the bottom part of the enclosure highlighting mounting pins

Mounting Pins

- Modelled mounting pin diameter: Ø 2.80mm
- Actual diameter of Arduino mounting pin: Ø 3.20mm

The smaller pin diameter allows light friction fit into the board instead of a press fit.

Mounting Pillar Placement

The following are the dimensions between the centres of the mounting pillars:

- Horizontal distance between mounting pillar centres: 66.10mm
- Distance between the tops of the top and bottom mounting pillars: 50.70mm

The remaining positional offsets describe where the pillars lie relative to the enclosure reference edges:

- Offsets: 14.00 mm, 15.30mm
- Vertical offset from lower reference edge: 2.50mm

These values match those of the Arduino Uno mounting pattern in the design tutorial.

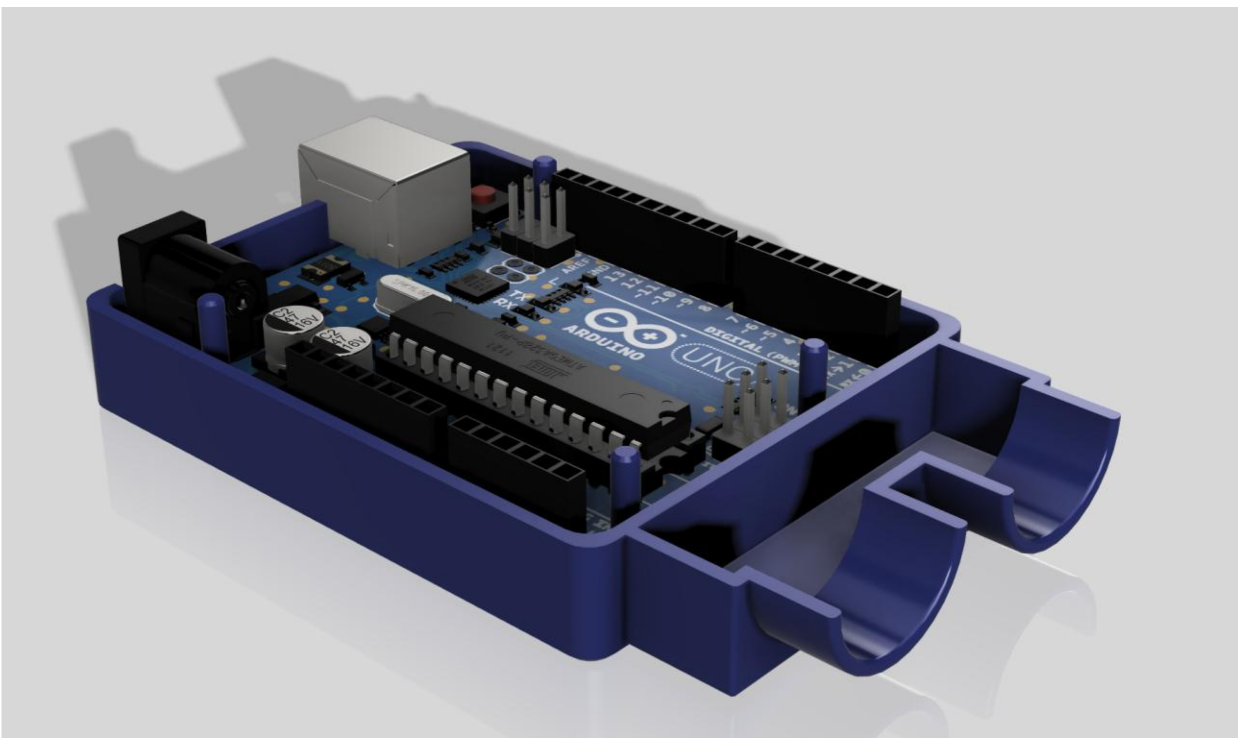

Figure 7: 3D model of the bottom part of the enclosure highlighting Arduino's position on it

Standoffs and Structural Support

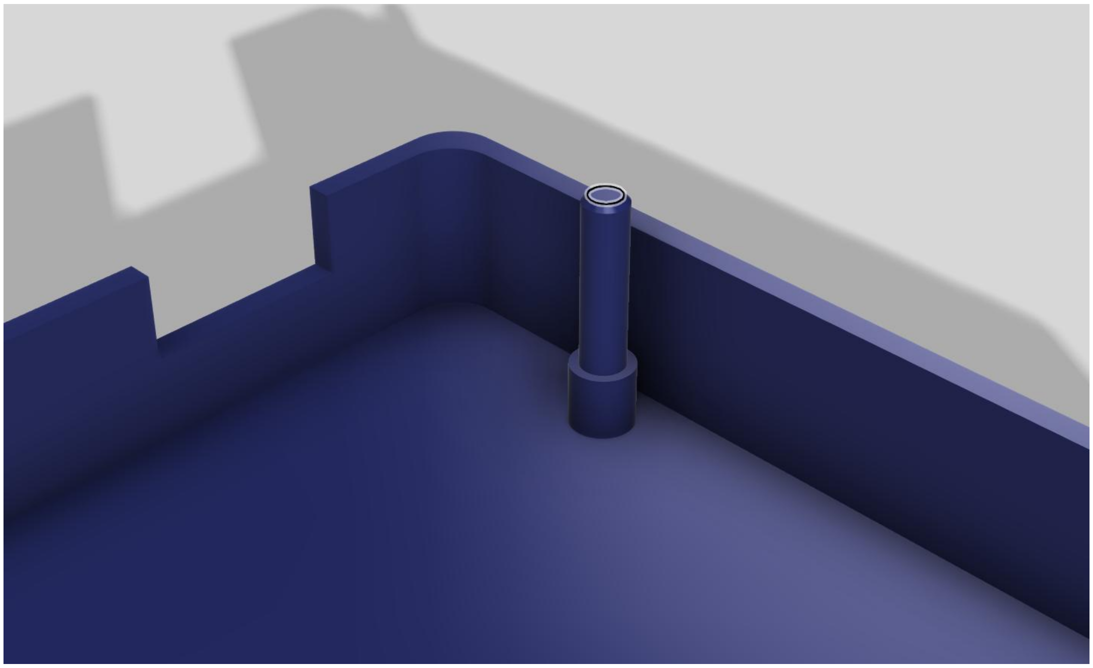

Figure 8: 3D model of the enclosure highlighting Standoffs

To prevent the Arduino PCB from touching the bottom of the enclosure (the solder joints and components on the board's underside), cylindrical standoffs were used.

- Standoff diameter: 4.0mm
- Standoff height: 4.0mm

These standoffs have the purpose of supporting the Arduino and providing stable alignment with cut-outs.

Top and Bottom Assembly Features

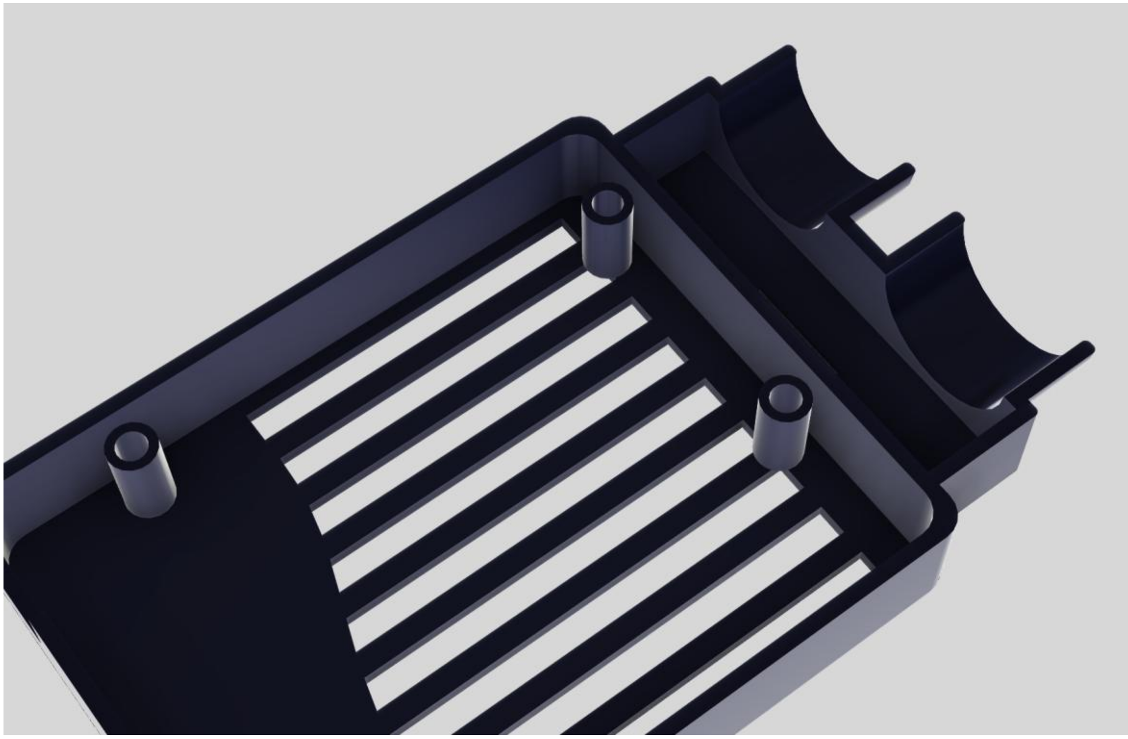

Figure 9: 3D model of the enclosure highlighting mounting pin receiving hole

The enclosure employs an interference-assisted snap-fit design:

- Pin overlap beyond the mounting split plane: 4.0mm
- Receiving hole diameter (top half): 5.2mm
- Receiving hole depth: 10.0mm
- Chamfer on mounting pins: 4.0mm

Chamfers were also added to help guide the pins into the receiving holes, making assembly easier and preventing damage to the prints in the process.

Ultrasonic Sensor Housing Integration

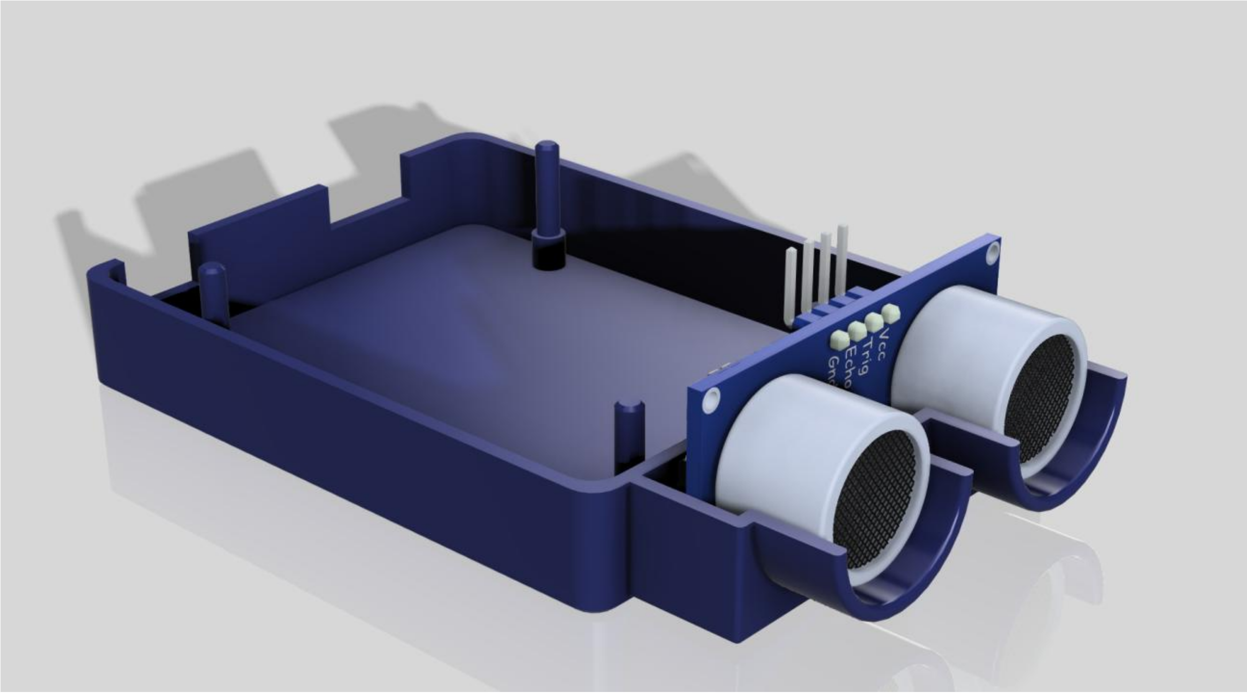

Figure 10: 3D model of the enclosure highlighting Ultrasonic Sensor Housing Integration

An extension was designed to install the HC-SR04 ultrasonic sensor in front of the enclosure's body, aligned with it.

The measured dimensions of the ultrasonic sensor housing are as follows:

- Front extension length from the main body: 12.621mm
- Sensor barrel insertion depth: 9.02mm
- Front bezel / ring thickness: 3.625mm

Two circular holes are provided that are aligned with the ultrasonic transmitter and receiver so that no obstructions would interfere its operation. This also maintains the geometry of the sensor with respect to the enclosure body.

Ventilation and Access Features

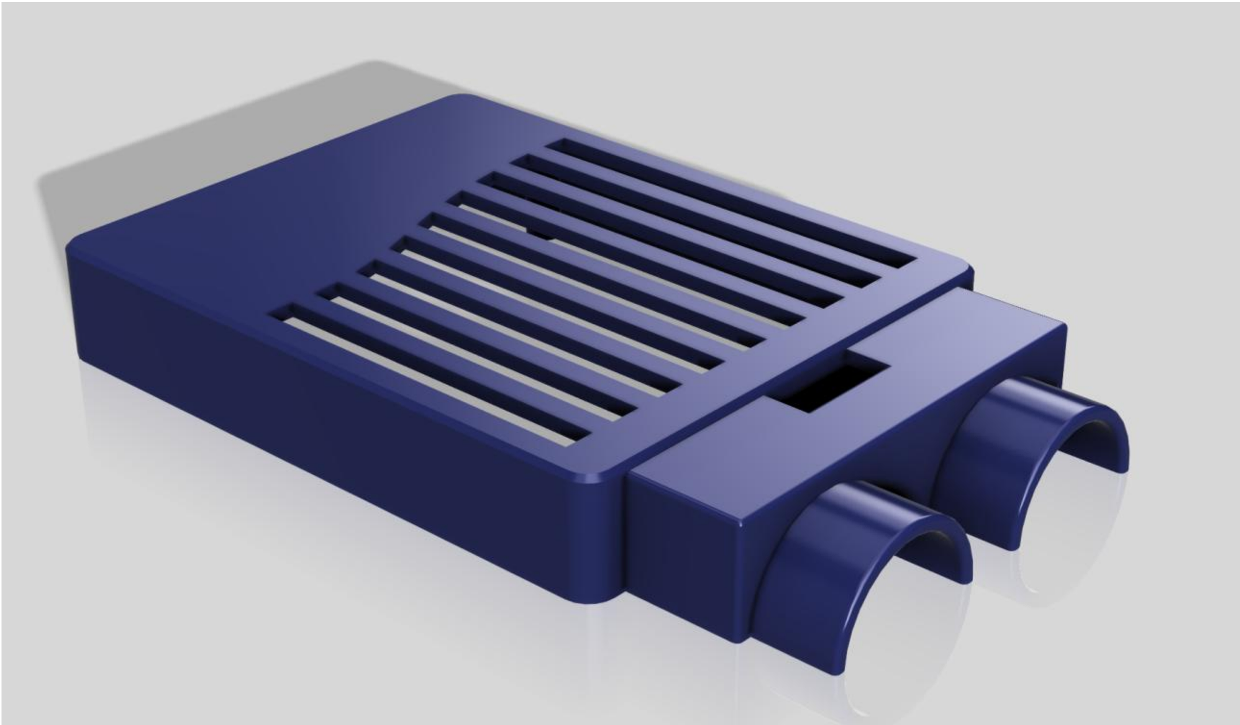

Figure 11: 3D model of the enclosure highlighting Ventilation and Access Features

To allow airflow to prevent overheating during operation, ventilation slots were also cut into the top cover:

- Slot opening width: 3.5mm
- Spacer width between slots: 2.5625mm
- Number of openings: 9
- Total vent length about 52mm

Extra cut-outs on the top surface allow direct access to the Arduino's programming and debug header pins without having to remove the lid of the enclosure.

Design Considerations and Manufacturability

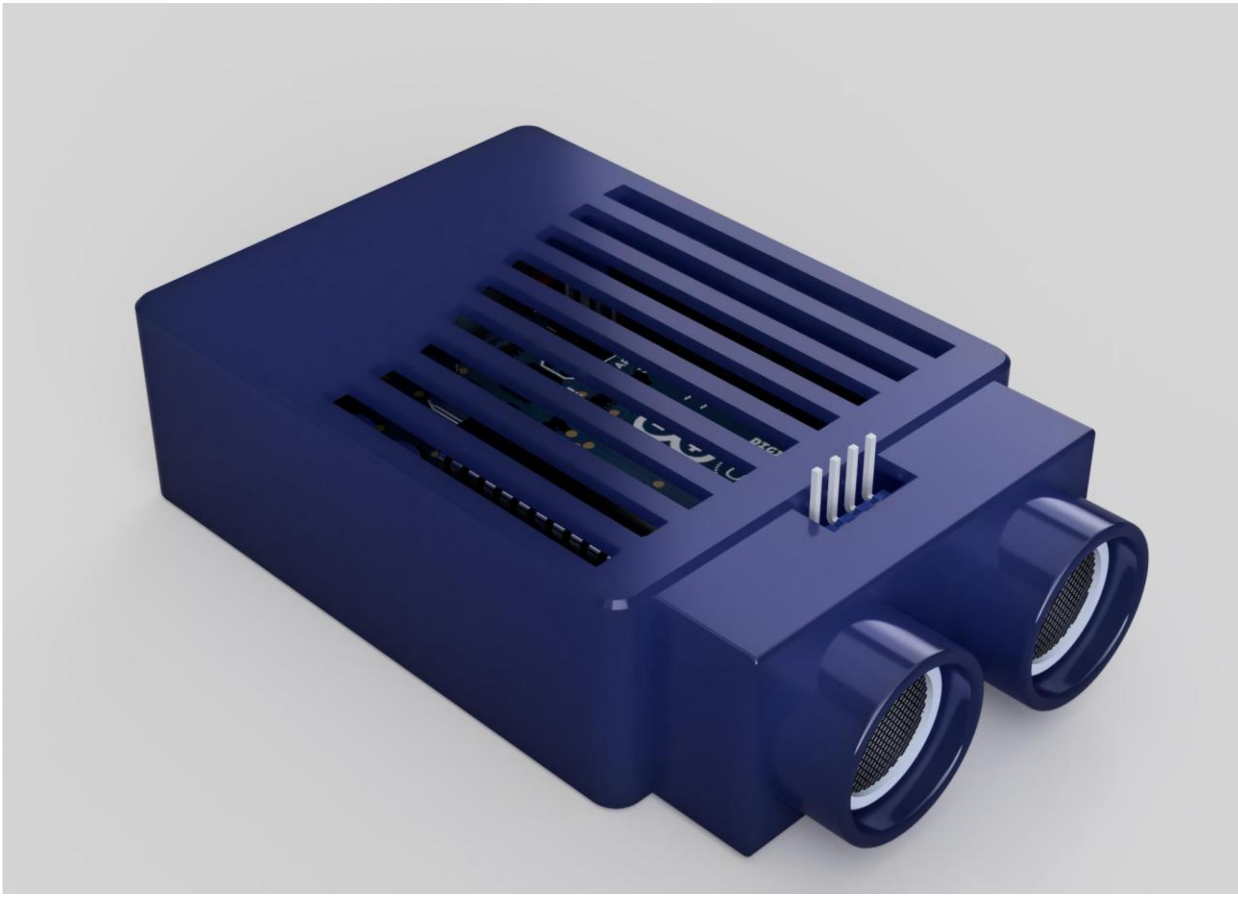

Figure 12: 3D model of the enclosure

The design of the enclosure was performed for additive manufacturing, using consistent wall thickness, large gaps, and chamfered edges to ensure a good result from the 3D printer and ease of assembly. The quantity of material was minimised to meet environmental targets while retaining required strength.

Enclosure Design Summary

The enclosure was based on the mounting geometry of the Arduino Uno with 1.0 mm internal clearance, a 1.6 mm wall thickness and an integrated housing for an ultrasonic sensor for accurate alignment, ease of assembly, and manufacturability.

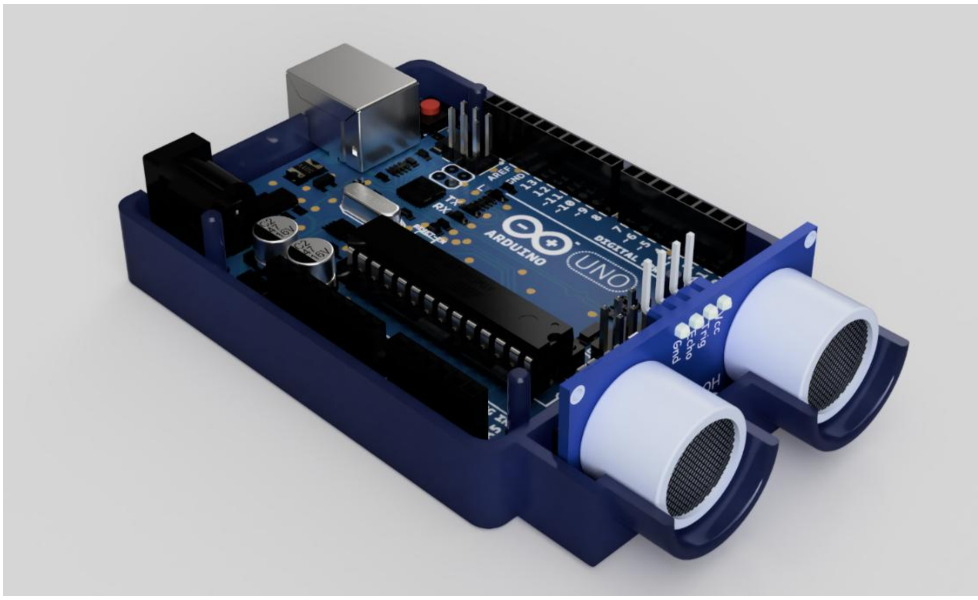

Figure 13: Enclosure design 1 ~ Appearance without cover - Blue

Figure 14: Enclosure design1 ~ Appearance with cover - Blue

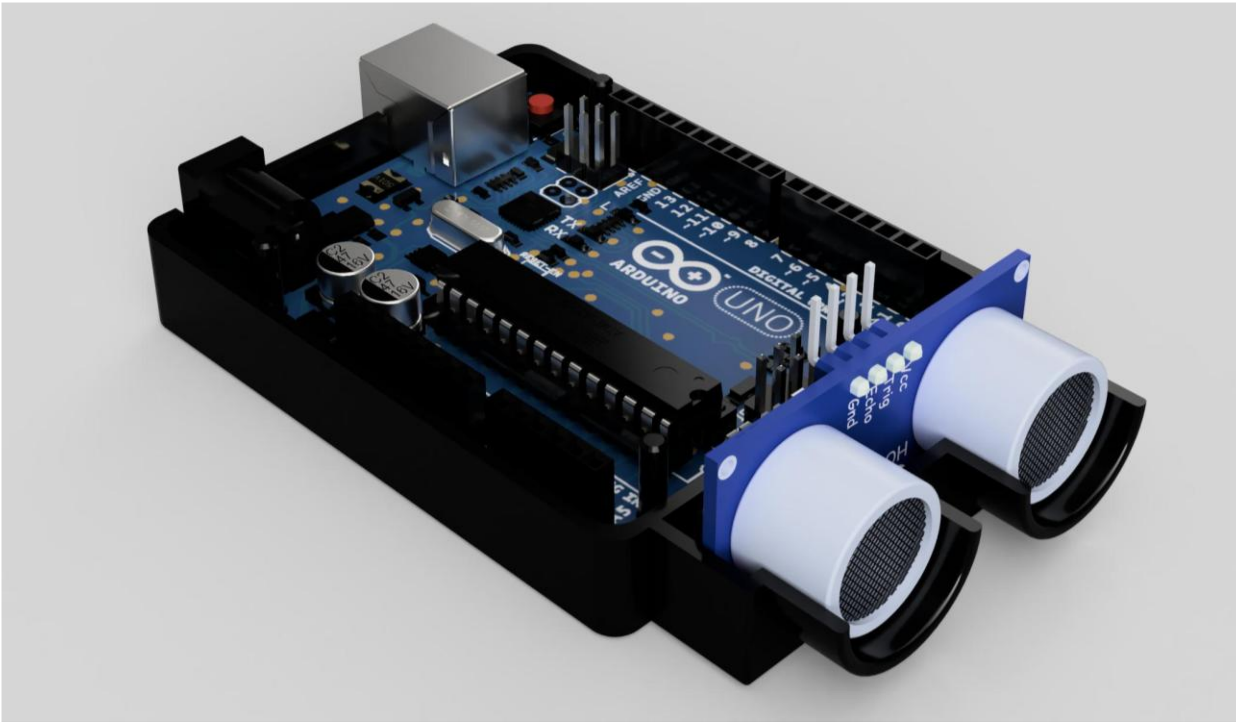

Figure 15: Enclosure design 2 ~ Appearance without cover - Clear

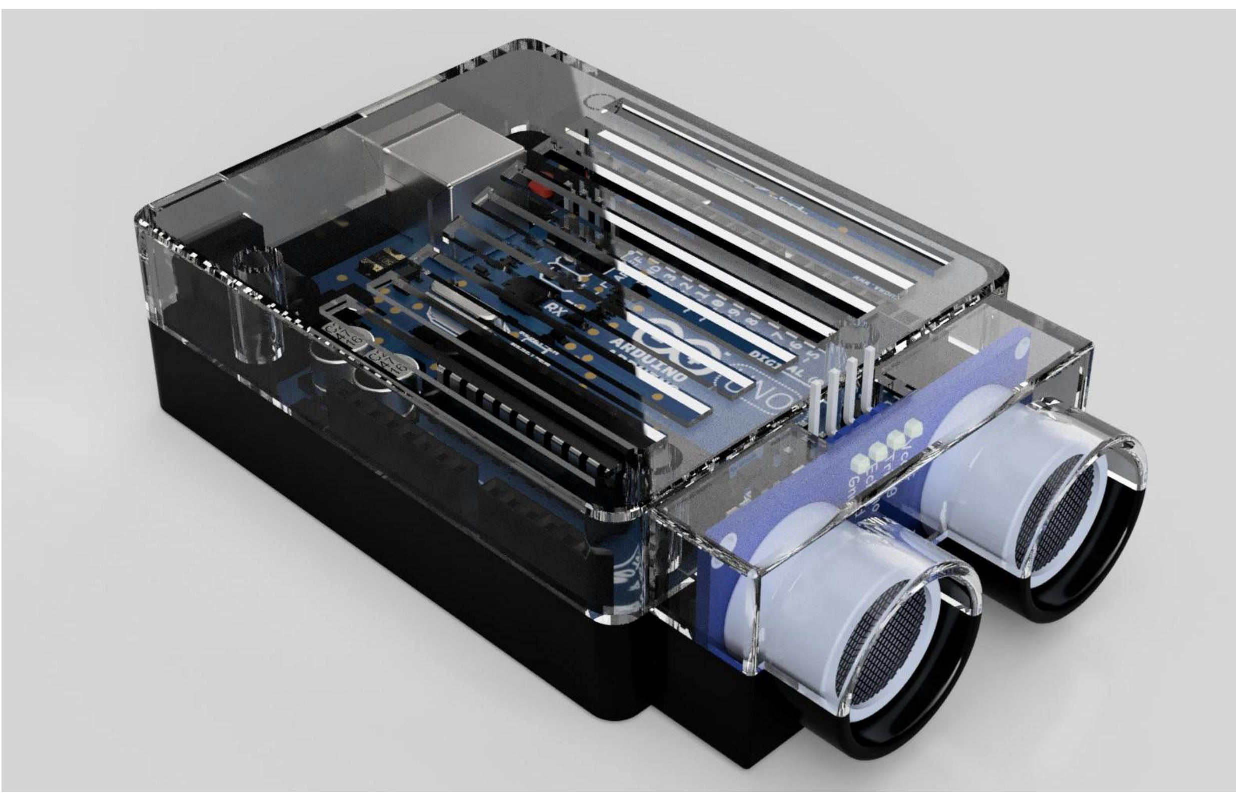

Figure 16: Enclosure design 2 ~ Appearance with cover - Clear

7. Customer Feedback

To understand how the product would be used in real- life situations we got the customers feedback on our device. We collected responses from a variety of people including professionals in fields such as Construction, Manufacturing and quality control, as well as people who take on advances home improvement projects. These results allowed us to understand how different users might interact with the project.

As we did before in the customer input, we created a questionnaire for people who have interacted with the device. The feedback showed a range of experience levels, with professionals generally having more familiarity with crack detection that home users. Nonetheless, similar views and priorities were recognised across both groups. Many users mentioned the ease of use, good design, reliability and practical features that make the device suitable for the desired fields. In addition, portion of respondents also said they would like the data to be easy to store in the device in cases of multiple checks in one area and more detailed output for easier reporting.

This feedback helped to understand the product reflects the needs and expectation of its intended users. It helped to identify the most important features to users. With the customer feedback we would be able to update our product in accordance to the needs and wants of the target audience.

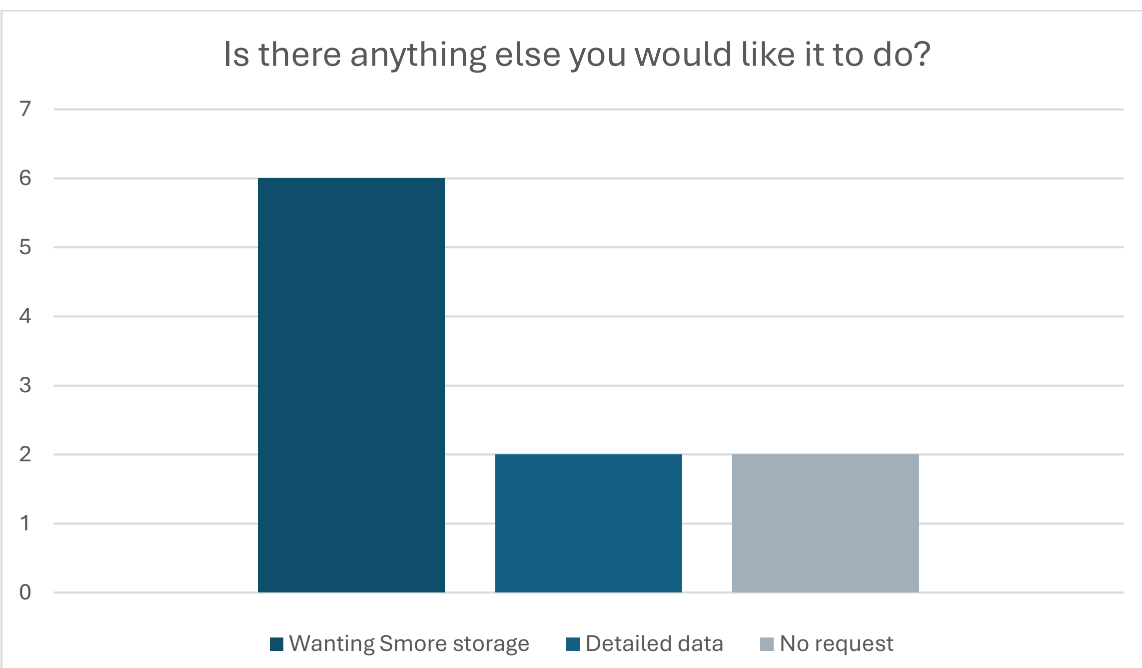

Figure 17: Desired Features in a Crack Detector from Survey

8. CONCLUSION

As part of this project, background research, customer survey, sensor experiments and prototypes have been undertaken, in keeping with United Nations, Sustainable Development Goal 9: Industry, Innovation, and Infrastructure, to show that early crack detection is a relevant social issue and technically feasible using low-cost hardware.

The pre-development survey identified that a large proportion of people are aware of cracks but do not have the knowledge or equipment to effectively assess their severity. Cost and portability were identified as two of the main barriers to the uptake of current crack detection technologies. The research also helped define the impact on the design of the resulting prototype which needed to be simple, low cost, and easy to use. This simplicity was confirmed by the results of the post-development feedback which said that the device looked practical, reliable and suitable for inspections.

The HC-SR04 ultrasonic sensor was tested, and it was found that the accuracy of its readings was well within limits determined by its datasheet. These limits are especially valid at small distances such as those it would be used at for crack detection on the surface. This establishes ultrasonic sensing as a viable option for the crack detection system. Additionally, a custom enclosure design has been created to fit the Arduino Uno and ultrasonic sensor with more compact protection, with an emphasis on correct sensor alignment, assembly and manufacture.

The prototype could possibly also find use in profiling structures and infrastructure, as well as detecting the early stages of structural failure. Although not a replacement for specialist structural analysis equipment, it could be used as a simple and potentially powerful early warning system against possible faults in a given structure. This could help prevent small structural cracks developing into failure.

Further work can be done in improving the accuracy of crack detection through more precise sensors or different sensing technologies. Other improvements include classification of crack severity, increased data storage, and the addition of wireless communication. Improvements to the enclosure and environmental sealing may allow for deployment of the system outdoors and long-term usage.

In conclusion, this project shows that a low-cost, simple crack detection device can be developed and employed to improve infrastructure safety and longevity in a more sustainable way.

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APPENDIX

Appendix A: Pre-Development Survey Responses

Survey Response 1

1. Are you aware of any technology or device that detects cracks?
   No, I have never used a device like that.
2. Would you trust a device to detect cracks?
   Yes, I've used devices like stud detectors before, and they work well.
3. Which of these features would you like in such a device?
   a. Crack severity classification (minor, moderate, severe)
   b. Record and store data from session to playback later
   c. Low cost
   d. Portable device
   e. Continuous monitoring
   f. Other (Specify)

   c - portable device, d - low cost
4. How likely would you be to use this device?
   I probably wouldn't use it. I usually just fill cracks at home without giving it too much thought so I wouldn't really be interested in getting one.
5. Where would you use this device?
   I'm not in construction or anything like that by profession so I would probably only use it at home if I had to get one.
6. Any other comments/suggestions?
   How much would a device like this usually cost?
   Responded: The lowest price I've seen is about €100.

Survey Response 2

1. Are you aware of any technology or device that detects cracks?
   Yes, but I have never used one before.

2. Would you trust a device to detect cracks?

   Yes, especially when it's not easy to see, like inside a material.

3. Which of these features would you like in such a device?
   a. Crack severity classification (minor, moderate, severe)
   b. Record and store data from session to playback later
   c. Low cost
   d. Portable device
   e. Continuous monitoring
   f. Other (Specify)

   a – severity classification, c – low cost, d – portable, e – continuous monitoring

4. How likely would you be to use this device?

   Yes, it would be useful for work.

5. Where would you use this device?

   I would use it at work my workplace and at home for personal projects.

6. Any other comments/suggestions?

   No

Survey Response 3

1. Are you aware of any technology or device that detects cracks?

   Yes, I've used tech like that.

2. Would you trust a device to detect cracks?

   Yes, technology is always more reliable than people.

3. Which of these features would you like in such a device?
   a. Crack severity classification (minor, moderate, severe)
   b. Record and store data from session to playback later
   c. Low cost
   d. Portable device
   e. Continuous monitoring
   f. Other (Specify)

   d – portable, f – Other: highly accurate and reliable

4. How likely would you be to use this device?

   Very likely, but only if I found it better than the equipment I use now.

5. Where would you use this device?

   I work in quality assurance, so I use tools like this at work.

6. Any other comments/suggestions?
7. ***It's very useful to be able to see exactly where a fault is early in production.***

Survey Response 4

1. Are you aware of any technology or device that detects cracks?
   Yes, I know of a couple tools that can detect cracks.
2. Would you trust a device to detect cracks?
   Yes, manually inspections can be really time consuming.
3. Which of these features would you like in such a device?
   a. Crack severity classification (minor, moderate, severe)
   b. Record and store data from session to playback later
   c. Low cost
   d. Portable device
   e. Continuous monitoring
   f. Other (Specify)

   a – classification, c – low cost, d - portable
4. How likely would you be to use this device?
   It's something I would like to have and not need rather than need and not have.
5. Where would you use this device?
   Probably at home and at work.
6. Any other comments/suggestions?
   No.

Survey Response 5

1. Are you aware of any technology or device that detects cracks?
   Yes, but I don't have one.
2. Would you trust a device to detect cracks?
   Yes, as long as it's decently accurate.
3. Which of these features would you like in such a device?
   a. Crack severity classification (minor, moderate, severe)
   b. Record and store data from session to playback later
   c. Low cost
   d. Portable device
   e. Continuous monitoring
   f. Other (Specify)

   c – low cost, d - portable
4. How likely would you be to use this device?
   I would use it if I had one.
   Responded: Why don't you have one?
   I've looked at a few devices, but they were a bit out of my price range.
5. Where would you use this device?

I would use it at work.

6. Any other comments/suggestions?

   No.

Appendix B: Post-Development Survey Responses

Survey Response 1

1. Does this device look like something you would use?

   Yes, it looks like something I would use. It seems sturdy and designed for practical work.
2. Is there anything you would change in the design?

   Maybe make add a handle so it's easy to carry around during jobs.
3. Does the device work the way you expect it to?

   Yes, it works pretty much how I expected it to.
4. Is there anything else you would want it to do?

   Saving previous results would be useful when checking the same area more than once.
5. How likely would you be to use this device?

   I'd be fairly likely to use it when needed.
6. Any other comments/feedback?

   It seems reliable for hands-on use.

Survey Response 2

1. Does this device look like something you would use?

   Yes, especially for inspections or assessments.
2. Is there anything you would change in the design?

   Yes, I would make curve in the side for better grip.
3. Does the device work the way you expect it to?

   Mostly yes, once you understand how to position it properly.
4. Is there anything else you would want it to do?

   Being able to store and export data would be useful.
5. How likely would you be to use this device?

   I would probably use it fairly regularly.
6. Any other comments/feedback?

   It feels like something designed for professional environments.

Survey Response 3

1. Does this device look like something you would use?

Yes, I could see myself using it.

2. Is there anything you would change in the design?
   Not much, the design makes sense for field use.
3. Does the device work the way you expect it to?
   Yes, it does what it's supposed to do.
4. Is there anything else you would want it to do?
   Nothing major, it already covers what I'd need.
5. How likely would you be to use this device?
   Probably now and then, depending on the project.
6. Any other comments/feedback?
   It looks durable enough for regular on-site use.

Survey response 4

1. Does this device look like something you would use?
   Possibly, *mainly for more detailed checks.*
2. Is there anything you would change in the design?
   Nope *it looks very industrial which suits.*
3. Does the device work the way you expect it to?
   It *works fine but there is a bit of a learning curve.*
4. Is there anything else you would want it to do?
   A *more detailed feedback (from the device) would be helpful for reporting.*
5. How likely would you be to use this device?
   Not *all the time, but useful in specific cases.*
6. Any other comments/feedback?
   No.

Survey Response 5

1. Does this device look like something you would use?
   Yes, definitely. It looks reliable and well suited for repeat checks.
2. Is there anything you would change in the design?
   No major changes – it feels well thought out.
3. Does the device work the way you expect it to?
   Yes, it performed as expected.
4. Is there anything else you would want it to do?
   Tracking results over time would be useful.
5. How likely would you be to use this device?
   Very likely, especially when monitoring the same areas.

6. Any other comments/feedback?
   It looks like something that would fit well into regular professional workflows.

Appendix C: Code

Ultrasonic sensor code

```
// C++ code
int trig = 3;
int echo = 2;

long duration; //For echo reading
float distance; //For ultrasonic sensor

void setup()
{
  pinMode(trig, OUTPUT);
  pinMode(echo, INPUT);
  Serial.begin(9600);
}

void loop()
{
  distance = calcDistance();
  Serial.print("Distance = ");
  Serial.println(distance);
  delay(300); //Delay for 300 microsecond (3 seconds)
}

float calcDistance()
{
  digitalWrite(trig, LOW);
  delayMicroseconds(2);
  digitalWrite(trig, HIGH); //Set trig HIGH for 10 microseconds
  delayMicroseconds(10);
  digitalWrite(trig, LOW);
  duration = pulseIn (echo, HIGH); //Read echo
  distance = (duration * 0.034) / 2;
  return distance;
}
```